Functional Requirements Document – Task Management

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document (sub-module 02.1 Task Management, code `TASK`) · **Version:** 1.0 · **Status:** Draft **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026 **Author:** Jojo George (jojogeorge3344@gmail.com) **Based on:** BRD-Task-Management.md v1.0 © 2026 Jojo George. All rights reserved.

Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version, derived from BRD-Task-Management v1.0 and the verified Task Management and Comments & Activity History workflow docs

Related Documents

Document	Purpose	Location
BRD – Task Management	Business needs this FRD details (`REQ-TASK-nn`, `BR-TASK-nn`)	`../../../BRD/02-TaskFlow-Monitoring/02.1-Task-Management/BRD-Task-Management.md`
Overall BRD / FRD	Roles, global rules `BR-nn`, global NFRs, Authentication and Legal	`../../../BRD/00-Overall/BRD-TaskFlow-Overall.md`, `../../00-Overall/FRD-TaskFlow-Overall.md`
Sibling FRDs – Employee Workload, Board	Screens that open or share task data	`../02.2-Employee-Workload/FRD-Employee-Workload.md`, `../02.3-Board/FRD-Board.md`
Development Plan	Screen-wise build plan (original Screens 7, 8, 9, 11, 15, 18)	`TF - Documents/Screens Wise Feature Development Plan/`
HLD / LLD / TDD	Technical design	`../../../HLD/02-TaskFlow-Monitoring/02.1-Task-Management/HLD-Task-Management.md`, `../../../LLD/02-TaskFlow-Monitoring/02.1-Task-Management/LLD-Task-Management.md`, `TDD-TaskFlow.md` (v2.0)

1. Introduction

1.1 Purpose

This FRD specifies the detailed, testable behaviour of the Task Management sub-module of TaskFlow Monitoring: the Task List, Create Task (including Copy), Task Details, Reassign Task, Task Discussion and Activity History screens. For each screen it states the inputs, outputs, rules, validations, messages and permissions, and traces every requirement back to the Task Management BRD.

1.2 Scope

In scope: the six screens above and the behaviour behind them (task scope, status rules, worked hours, soft delete, comments, activity history and the notifications and internal mail these actions send).

Out of scope (see BRD §3.2): choosing an initial status on Create; editing reminder intervals after creation; a screen for changing additional recipients after creation; the Board (`../02.3-Board/FRD-Board.md`); Employee Workload (`../02.2-Employee-Workload/FRD-Employee-Workload.md`); sending deadline reminders (Notification, `../../04-Communication/04.2-Notification/FRD-Notification.md`); task reports (`../../05-Reports/05.1-Reports/FRD-Reports.md`).

1.3 Audience

Developer, tester and reviewer of TaskFlow (Jojo George), and the product owner approving the behaviour.

1.4 Conventions

- Requirement IDs: `FR-TASK-nn` . Use cases: `UC-TASK-nn` . BRD requirements: `REQ-TASK-nn` ; sub-module rules `BR-TASK-nn` ; global rules `BR-nn` (overall BRD).
- Notification codes in §7: `N-TASK-nn` .
- Priority: **Must / Should / Could** (MoSCoW).
- "The system shall …" = mandatory behaviour; "should" = recommended.
- Text in quotes is the exact UI label or message; `{name}` marks a value filled in at run time.

1.5 Definitions & Abbreviations

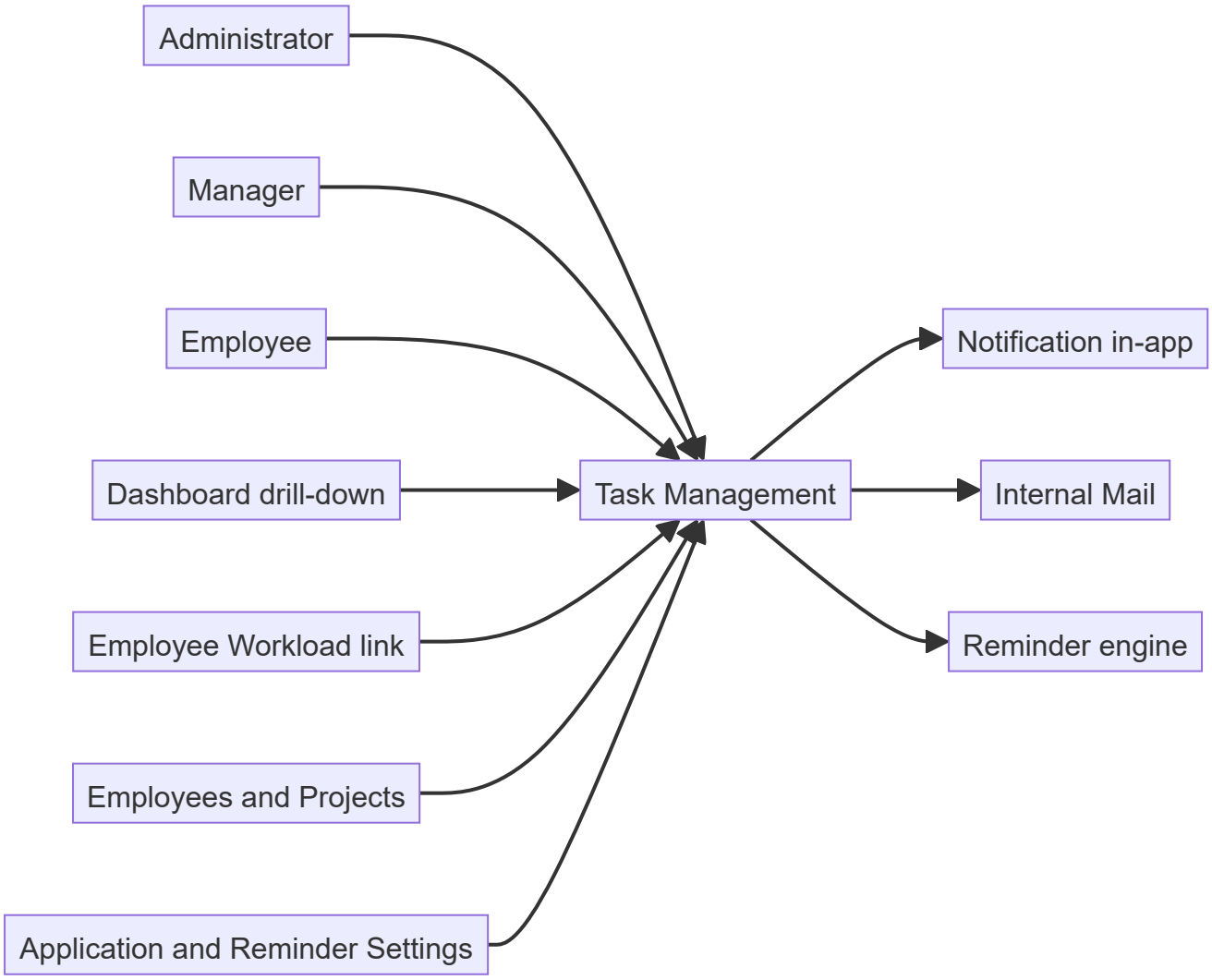
Term	Meaning
Task scope	The tasks a user may see: **Own** (assigned, created or notified), **Team** (Manager) or **Organization** (Administrator)
Team (Manager)	Active people in a department the Manager manages or on a project where the Manager is flagged as project manager
Recipient	A user notified about a task; the assignee is always one
Stage	Upcoming, Due today, Overdue or Completed – worked out from the deadline and status each time a task is shown
Task code	`TSK-` followed by the 6-digit task number, e.g. `TSK-000042`
Worked hours / Remaining	Hours logged against the task / estimated hours minus worked hours
Soft delete	The record is hidden from every list but kept, with its comments and history
Drill-down	Opening the Task List from a dashboard chart segment, showing exactly the tasks it counted
Internal mail	A message in the TaskFlow Mail module (no external e-mail, global BR-23)

2. System Overview

2.1 Module Map

Code	Module	Child modules / screens	Primary roles
TASK	Task Management (TaskFlow Monitoring)	Task List, Create Task (and Copy), Task Details, Reassign Task	Administrator, Manager, Employee, Custom role
TASK	Comments & Activity History (part of Task Management)	Task Discussion, Activity History	All roles with task access
BOARD	Board (related)	Board (Kanban) – shares task scope, status rules and history	All roles with `board.view`
WORK	Employee Workload (related)	**Tasks** → link opens the Task List for one employee	Administrator, Manager
NOTIF / MAIL	Notification, Mail (related)	Receive task notifications and internal mail; reminder engine	All users
DASH	Dashboard (related)	Chart segments open the Task List as a drill-down	All users

2.2 Context Diagram



3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Administrator	System owner; every task action including delete and worked-hours corrections	Organization (all tasks)
Manager	Leads a team; creates, copies, edits and reassigns tasks; changes status; comments	Team, plus tasks the Manager created, is assigned or is notified about
Employee	Works on tasks; views own tasks, changes status, logs hours on own tasks, comments	Own (assigned, created or notified)
Custom role	Role defined in Role Configuration	Own by default; Team or Organization when `tasks.view.all` is granted

3.2 Permission Matrix (functional view)

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Task List › View my tasks (list, board, details)	`tasks.view.own`	Always On	Always On	Always On	Always On
Task List › View all tasks (Managers limited to their team)	`tasks.view.all`	Granted	Granted	Denied	Configurable
Task List › Create and copy tasks	`tasks.create`	Granted	Granted	Denied	Configurable
Task List › Delete tasks	`tasks.delete`	Granted	Denied	Denied	Configurable
Details › Edit tasks (and log hours on any accessible task)	`tasks.update`	Granted	Granted	Denied	Configurable
Details › Reassign tasks	`tasks.reassign`	Granted	Granted	Denied	Configurable
Details › Manage task recipients (no screen in this release)	`tasks.recipients.manage`	Granted	Granted	Denied	Configurable
Details › Update task status (shared with Board)	`tasks.status.change`	Granted	Granted	Granted	Configurable
Discussion › Post task comments	`tasks.comment.create`	Granted	Granted	Granted	Configurable
Discussion › Edit own task comments	`tasks.comment.edit`	Granted	Granted	Granted	Configurable
Discussion › Delete own task comments	`tasks.comment.delete`	Granted	Granted	Granted	Configurable
Activity › View task history	`tasks.activity.view`	Granted	Granted	Granted	Configurable
Activity › Correct or delete a worked-hours entry	none – Administrator role only	Granted	Denied	Denied	Denied

Supporting keys used by these screens (owned by other sub-modules): `employees.view` and `projects.view` (assignee, recipient and project lists on Create and Reassign) and `settings.reminders.manage` (reminder defaults

on Create) – Administrator and Manager Granted, Employee Denied, Custom role Configurable.

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4. Functional Requirements by Module

4.1 Module: Task Management (TASK)

Purpose: Keep one reliable record per task – owner, deadline, status, effort, discussion and history – and let authorised users act on it. · **BRD reference:** BRD-Task-Management §5 SCREENS 1–6, BR-TASK-01…12; global BR-06…BR-13, BR-17…BR-19, BR-21…BR-23.

4.1.1 Screen: Task List

Field	Value
Route / location	`/tasks` – menu **TaskFlow Monitoring** → **Task Management** (with `tasks.view.all`) or **Task Management** → **My Tasks** (everyone else)
Access	All signed-in users (`tasks.view.own`, Always On); content follows the user's task scope
Entry points	Menu; dashboard chart segment (drill-down); **Tasks** → on Employee Workload; Back from Task Details

Screen elements

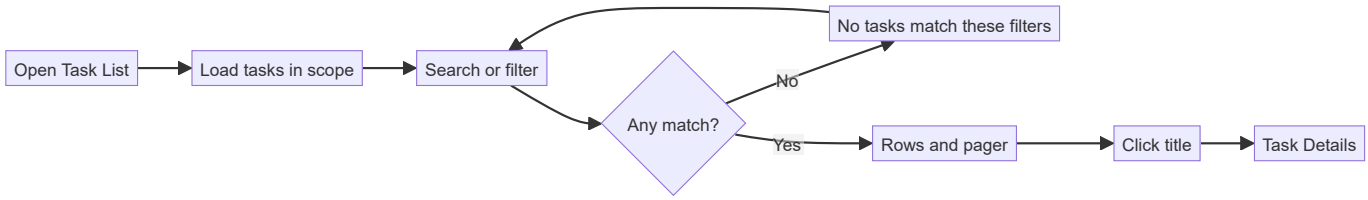
Element	Type	Description	Mandatory	Visible to
Count text	Text	"1–20 of 57 tasks", plus " match the current filters" when filters are active	–	All
Board view	Button	Opens the Board	–	All (see §10, open issue)
Create task	Button	Opens Create Task	–	`tasks.create`
Search	Text input	Placeholder "Task code or title"	No	All
Employee	Drop-down	Filter by assignee	No	`tasks.view.all`
Project	Drop-down	Filter by project (shown when the project list loads)	No	All
Status, Priority	Drop-downs	Filter by status / priority	No	All
Deadline from, Deadline to	Date inputs	Deadline range	No	All
Reset	Button	Clears all filters	–	All
Drill-down banner	Banner	"Opened from the dashboard: …" with **Show all tasks**	–	When opened from a dashboard segment
Task table	Table	Code, Title (link), Project, Assignee, Creator, Priority, Start, Deadline (stage chip), Est. hours, Status	–	All
Copy task	Row action	Opens Create Task prefilled from the row	–	`tasks.create`
Delete	Row action	Deletes the task after confirmation	–	`tasks.delete`
Pager	Buttons / text	Previous, Next, "Page X of Y"	–	All

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-01	The system shall list only the tasks in the user's task scope: Own (assigned, created or notified) for users without `tasks.view.all`, Team plus own for a Manager, Organization for an Administrator.	Must	`tasks.view.own`, `tasks.view.all`	REQ-TASK-01, BR-TASK-01, BR-06
FR-TASK-02	The system shall define a Manager's team as the active people in a department the Manager manages or on a project where the Manager is flagged as project manager.	Must	`tasks.view.all`	BR-TASK-01
FR-TASK-03	The system shall order the list by deadline, soonest first, and show 20 tasks per page (at most 100 per page) with Previous, Next and "Page X of Y".	Must	`tasks.view.own`	REQ-TASK-02, NFR-TASK-01, NFR-01
FR-TASK-04	The system shall search task code and title and refresh the list 300 ms after the user stops typing.	Must	`tasks.view.own`	REQ-TASK-03
FR-TASK-05	The system shall filter by Status, Priority, Project and a Deadline from/to range, combine all chosen filters (AND) and restart at page 1 whenever a filter changes.	Must	`tasks.view.own`	REQ-TASK-04
FR-TASK-06	The system shall offer the Employee filter only to users with `tasks.view.all`.	Should	`tasks.view.all`	REQ-TASK-05
FR-TASK-07	The system shall show a stage chip "Overdue", "Due today" or "Upcoming" on each deadline and highlight overdue rows.	Must	`tasks.view.own`	REQ-TASK-06, BR-12
FR-TASK-08	The system shall, when opened from a dashboard chart segment, list exactly the tasks the segment counted, show the banner "Opened from the dashboard: …" and offer **Show all tasks** to return to the unfiltered list.	Should	`tasks.view.own`	REQ-TASK-07, BR-22
FR-TASK-09	The system shall limit a Manager's drill-down list to the team's tasks.	Must	`tasks.view.all`	BR-TASK-02
FR-TASK-10	The system shall open Task Details when the user clicks a task title.	Must	`tasks.view.own`	REQ-TASK-08
FR-TASK-11	The system shall show **Create task** and **Copy task** only to users with `tasks.create`, and **Delete** only to users with `tasks.delete`.	Must	`tasks.create`, `tasks.delete`	REQ-TASK-09
FR-TASK-12	The system shall title the page "Task Management" for users with `tasks.view.all` and "My Tasks" for everyone else.	Could	`tasks.view.all`	REQ-TASK-10
FR-TASK-13	The system shall show "No tasks match these filters" with **Clear filters** when filters return nothing, "No tasks yet" / "Tasks assigned to you will appear here." when the user has no tasks, and "Could not load tasks. Check your connection and try again." when loading fails.	Should	`tasks.view.own`	REQ-TASK-02, REQ-TASK-04

Use case: UC-TASK-01 – Find and open a task

Item	Detail
Actor	Employee, Manager or Administrator
Preconditions	Signed in (two-step verification done)
Trigger	Opens **My Tasks** or **TaskFlow Monitoring → Task Management**
Main flow	1. System loads the first 20 tasks in the user's scope, soonest deadline first. 2. User types a code or title in **Search**; the list refreshes 300 ms after typing stops. 3. User picks Status, Priority, Project, a deadline range (and Employee if allowed); the list restarts at page 1. 4. User pages with Previous / Next. 5. User clicks a title; Task Details opens.
Alternate flows	A1: Opened from a dashboard segment – the list shows the segment's tasks with the drill-down banner; **Show all tasks** clears it. A2: Opened from Employee Workload **Tasks** → – the list is filtered to that employee within the user's scope.
Exceptions	E1: No match → "No tasks match these filters" and **Clear filters**. E2: Load failure → "Could not load tasks. Check your connection and try again."
Postconditions	The user sees the tasks they are allowed to see; nothing is changed



Use case: UC-TASK-02 – Open a dashboard drill-down

Item	Detail
Actor	Any user with a dashboard
Preconditions	Signed in; on a dashboard chart
Trigger	Clicks a chart segment
Main flow	1. System opens `/tasks` with the segment's criteria. 2. System lists exactly the counted tasks (team tasks only for a Manager) and shows "Opened from the dashboard: …". 3. User clicks **Show all tasks** to clear the drill-down.
Alternate flows	–
Exceptions	E1: Segment dates reversed → "The end date cannot be before the start date."
Postconditions	The list matches the chart count

Field validations

Field	Rule	Error message
Deadline from / to (drill-down)	End not before start	"The end date cannot be before the start date."
Search	Free text, code or title	–

Business rules applied: BR-TASK-01, BR-TASK-02, BR-TASK-04; global BR-06, BR-12, BR-17, BR-22.

Notifications / side effects

Event	Who is notified	Channel
–	Viewing and filtering send nothing	–

Messages

Situation	Message
Load failed	"Could not load tasks. Check your connection and try again."
No result for filters	"No tasks match these filters" (button **Clear filters**)
No tasks at all	"No tasks yet" / "Tasks assigned to you will appear here."
No permission for an action	The button is hidden; a direct request is refused

4.1.2 Screen: Create Task (and Copy Task)

Field	Value
Route / location	`/tasks/new`; Copy opens `/tasks/new?copyFrom={id}`
Access	`tasks.create` (Administrator, Manager; Custom role as granted). Also needs `employees.view`, `projects.view` and `settings.reminders.manage` to fill the drop-downs and defaults
Entry points	Task List **Create task**; **Copy task** on a list row; **Copy** on Task Details

Screen elements

Element	Type	Description	Mandatory	Visible to
Project	Drop-down	Active projects	Yes	`tasks.create`
Title	Text input	Up to 200 characters	Yes	`tasks.create`
Description	Text area	Up to 4000 characters	No	`tasks.create`
Assigned employee	Drop-down	Active users with the Employee role	Yes	`tasks.create`
Priority	Selector	Low, Medium, High, Critical; default from Application Settings	Yes	`tasks.create`
Start date, Deadline	Date-time inputs	Default now and now + 3 days	Yes	`tasks.create`
Estimated hours	Number input	Greater than 0	Yes	`tasks.create`
Additional recipients	Checklist	Active users except the assignee; hint "The assigned employee is always notified."	No	`tasks.create`
First reminder, Second reminder	Number inputs	Days before the deadline (0–30), prefilled from Reminder Settings	No	`tasks.create`
After the deadline	Selector	Daily / Weekly, prefilled from Reminder Settings	No	`tasks.create`
Copy notice	Notice	"Copied from TSK-…. Check the dates before creating."	–	When opened with Copy
Cancel, Create task	Buttons	Leave without saving / save the task	–	`tasks.create`

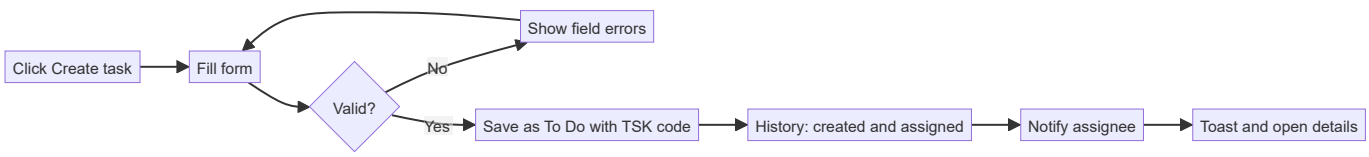
Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-14	The system shall give every new task a unique, never-reused code `TSK-` followed by the 6-digit task number (e.g. `TSK-000042`).	Must	`tasks.create`	REQ-TASK-11, BR-07
FR-TASK-15	The system shall require Project, Title, Assigned employee, Priority, Start date, Deadline and Estimated hours.	Must	`tasks.create`	REQ-TASK-12, BR-08
FR-TASK-16	The system shall reject a Title longer than 200 characters and a Description longer than 4000 characters.	Must	`tasks.create`	REQ-TASK-13
FR-TASK-17	The system shall offer the priorities Low, Medium, High and Critical and preselect the default priority from Application Settings.	Must	`tasks.create`	REQ-TASK-14
FR-TASK-18	The system shall preset Start date to the current date and time and Deadline to three days later.	Should	`tasks.create`	REQ-TASK-15
FR-TASK-19	The system shall reject a Deadline that is not later than the Start date.	Must	`tasks.create`	REQ-TASK-16, BR-08
FR-TASK-20	The system shall reject Estimated hours that are not greater than 0.	Must	`tasks.create`	REQ-TASK-17, BR-08
FR-TASK-21	The system shall save every new task with status To Do; the form has no status field.	Must	`tasks.create`	REQ-TASK-18, BR-11
FR-TASK-22	The system shall always add the assignee as a recipient, accept up to 50 additional recipients chosen from active users, and reject the task if any recipient does not exist.	Must	`tasks.create`	REQ-TASK-19, BR-TASK-05, BR-09
FR-TASK-23	The system shall accept First and Second reminder values of 0 to 30 days before the deadline (0 = deadline day) and an overdue frequency of Daily or Weekly, reject a second reminder that comes before the first, and use the Reminder Settings defaults for empty fields.	Must	`tasks.create`	REQ-TASK-20, BR-TASK-06, BR-14
FR-TASK-24	The system shall accept only an assignee who exists and is active and a project that exists and is active.	Must	`tasks.create`	REQ-TASK-21, BR-08
FR-TASK-25	The system shall list only active users with the Employee role in the Assigned employee drop-down.	Must	`tasks.create`, `employees.view`	REQ-TASK-22
FR-TASK-26	The system shall record the creator and creation time and add the history entries "Task created" and "Assigned to {name}".	Must	`tasks.create`	REQ-TASK-23, BR-19
FR-TASK-27	The system shall notify the assignee in-app with "A task was assigned to you: {title}" as part of saving the task.	Must	`tasks.create`	REQ-TASK-24, BR-09

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-28	The system shall, after saving, show "{code} created and assigned to {name}" and open the new task's details.	Should	`tasks.create`	REQ-TASK-11, REQ-TASK-24
FR-TASK-29	The system shall, when **Copy** is used, prefill the form from the source task (keeping the new-task date defaults), show "Copied from TSK-…. Check the dates before creating.", and save nothing until the user clicks **Create task**.	Should	`tasks.create`	REQ-TASK-25
FR-TASK-30	The system shall show "Copied from TSK-…, but {name} can no longer be assigned. Choose another employee." when the source assignee cannot be chosen, and "That task could not be copied, so the form is empty." when the source task cannot be loaded.	Should	`tasks.create`	REQ-TASK-25

Use case: UC-TASK-03 – Create and assign a task

Item	Detail
Actor	Manager or Administrator
Preconditions	Signed in; holds `tasks.create` (and the supporting view keys)
Trigger	Clicks **Create task** on the Task List
Main flow	1. System opens the form with the default priority, start now, deadline now + 3 days and the reminder defaults. 2. User picks Project, enters Title, picks Assigned employee, Priority, dates and Estimated hours, and optionally recipients and reminders. 3. User clicks **Create task**. 4. System validates the form and the server rules. 5. System saves the task as To Do with a `TSK-` code and adds the assignee as a recipient. 6. System records "Task created" and "Assigned to {name}" and notifies the assignee. 7. System shows "{code} created and assigned to {name}" and opens Task Details.
Alternate flows	A1: **Cancel** – nothing is saved.
Exceptions	E1: Missing or invalid field → field message (see validations); nothing saved. E2: Assignee or project inactive → "The assigned employee does not exist or is not active." / "The selected project does not exist." E3: No permission → "Your role is not allowed to create tasks."; without `tasks.create` the route sends the user back to the dashboard.
Postconditions	Task saved as To Do; assignee notified; history started; task visible to authorised users



Use case: UC-TASK-04 – Copy a task

Item	Detail
Actor	Manager or Administrator
Preconditions	Holds `tasks.create`; can access the source task
Trigger	Clicks **Copy task** on a list row or **Copy** on Task Details
Main flow	1. System opens Create Task filled from the source task, with new-task dates. 2. System shows "Copied from TSK-…. Check the dates before creating." 3. User adjusts the values and continues as UC-TASK-03 from step 3.
Alternate flows	A1: Source assignee can no longer be assigned → warning naming the person; user chooses another employee.
Exceptions	E1: Source cannot be loaded → "That task could not be copied, so the form is empty."
Postconditions	A new, independent task (the source is unchanged)

Field validations

Field	Rule	Error message
Project	Required; must exist and be active	"Project is required." · "The selected project does not exist."
Title	Required; at most 200 characters	"Title is required." · "Title must be 200 characters or fewer."
Description	At most 4000 characters	"Description must be 4000 characters or fewer."
Assigned employee	Required; must exist and be active	"Select the employee to assign." · "The assigned employee does not exist or is not active."
Start date	Required	"Start date is required."
Deadline	Required; later than Start date	"Deadline is required." · "Deadline must be after the start date."
Estimated hours	Required; greater than 0	"Estimated hours are required." · "Estimated hours must be a positive number."
Additional recipients	At most 50; every user must exist	"A task can have at most 50 additional recipients." · "One or more recipients do not exist."
First / Second reminder	0–30 days; second not before first	"The second reminder cannot come before the first reminder."

Business rules applied: BR-TASK-05, BR-TASK-06, BR-TASK-12; global BR-07, BR-08, BR-09, BR-10, BR-11, BR-14, BR-19.

Notifications / side effects

Event	Who is notified	Channel
Task created and assigned (N-TASK-01)	Assignee	In-app
Reminder intervals stored	Used later by the reminder engine for the recipients and creator	In-app and Mail (Notification sub-module)

Messages

Situation	Message
Success	"TSK-000123 created and assigned to {name}"
No permission	"Your role is not allowed to create tasks."
Copy notice	"Copied from TSK-…. Check the dates before creating."

4.1.3 Screen: Task Details

Field	Value
Route / location	`/tasks/{id}`
Access	Assignee, creator and recipients of the task; `tasks.view.all` holders (a Manager only for team tasks); Administrator. Each action has its own key
Entry points	Task title on the Task List; card on the Board (Back returns to the Board); links in notifications and mail

Screen elements

Element	Type	Description	Mandatory	Visible to
Header	Text / chip	Task code, project, stage chip	–	Task access
Copy, Delete, Edit task	Buttons	Copy the task / delete it / open the edit form	–	`tasks.create` / `tasks.delete` / `tasks.update`
Details card	Read-only fields	Description, project, assignee, created by, priority, start date, deadline, estimated hours, worked hours, status, last updated, "Also notified"	–	Task access
Edit form	Inputs	Title, Description, Priority, Start date, Deadline, Estimated hours, Worked hours; **Cancel** / **Save changes**	Title, dates, estimate	`tasks.update`
Discussion tab	Tab	Comment count, latest 3 comments, "Showing the latest 3 of N.", **Open full discussion** →	–	Task access
Activity history tab	Tab	Latest 4 entries, **Open full history** →	–	`tasks.activity.view`
Reassign task tab	Tab	**Choose new assignee**	–	`tasks.reassign`
Time tracking	Panel	Estimated, Worked, Remaining; **Log hours** box; "View hours in activity history"	–	Assignee or `tasks.update`
Move to	Buttons	One button per allowed next status	–	Task access (action needs `tasks.status.change`)

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-31	The system shall show the task code, title, description, project, assignee, creator, priority, start date, deadline, estimated hours, worked hours, status, last updated time and the "Also notified" recipients.	Must	`tasks.view.own`	REQ-TASK-26
FR-TASK-32	The system shall open a task only for users in its scope and refuse anyone else with "You do not have access to this task.".	Must	`tasks.view.own`, `tasks.view.all`	REQ-TASK-26, BR-TASK-01, NFR-TASK-02
FR-TASK-33	The system shall show the stage Upcoming, Due today, Overdue or Completed, worked out in the application time zone each time the task is shown and never stored.	Must	`tasks.view.own`	REQ-TASK-27, BR-TASK-04, BR-12, BR-17
FR-TASK-34	The system shall let users with `tasks.update` change title, description, priority, start date, deadline, estimated hours and worked hours, and show "Task updated" after saving.	Must	`tasks.update`	REQ-TASK-28
FR-TASK-35	The system shall validate the edit form: Title required and at most 200 characters, Start date and Deadline required with the deadline after the start, Estimated hours at least 0.5.	Must	`tasks.update`	REQ-TASK-28, REQ-TASK-16
FR-TASK-36	The system shall record "Deadline updated", "Priority updated" and "Worked hours logged: {X}h (total)" history entries when those values change on edit.	Must	`tasks.update`	REQ-TASK-29, BR-19
FR-TASK-37	The system shall offer under **Move to** only the statuses allowed from the current status by the transition table in BR-TASK-03.	Must	`tasks.status.change`	REQ-TASK-30, BR-TASK-03, BR-11
FR-TASK-38	The system shall refuse a status change without `tasks.status.change` or outside the transition table, with "Cannot move a task from {from} to {to}." for a disallowed move.	Must	`tasks.status.change`	REQ-TASK-30, BR-TASK-03
FR-TASK-39	The system shall, on an accepted status change, record "Status changed from {X} to {Y}" (plus "Task completed" when completed) and show "{code} moved to {status}".	Must	`tasks.status.change`	REQ-TASK-30, REQ-TASK-52
FR-TASK-40	The system shall store the completion time when a task becomes Completed and the cancel time when it becomes Cancelled.	Should	`tasks.status.change`	REQ-TASK-31
FR-TASK-41	The system shall let the assignee, or a user with `tasks.update`, log more than 0 and at most 24 hours per entry, add them to the worked total, record "Worked hours logged: {X}h (total)" and show "{X}h logged".	Must	none for the assignee, otherwise `tasks.update`	REQ-TASK-32, BR-TASK-07
FR-TASK-42	The system shall show Estimated, Worked and Remaining hours in Time tracking and update them after hours are logged.	Should	`tasks.view.own`	REQ-TASK-33
FR-TASK-43	The system shall notify the assignee and the recipients in-app and by internal mail whenever the worked hours change (Log hours or edit).	Should	–	REQ-TASK-34

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-44	The system shall refuse a worked-hours save that clashes with another save, keep the other save's hours and show "The task's worked hours changed while you were saving. Please try again.".	Should	–	REQ-TASK-35, NFR-TASK-05
FR-TASK-45	The system shall show the latest 3 comments in the Discussion tab and, to users with `tasks.activity.view`, the latest 4 history entries in the Activity history tab.	Should	`tasks.activity.view`	REQ-TASK-36
FR-TASK-46	The system shall let users with `tasks.delete` delete a task after the confirmation "Delete task?" ("… will be removed for everyone. Its comments and history are kept."), hide it from every list, keep its comments and history, record "Task deleted" and show "{code} deleted".	Must	`tasks.delete`	REQ-TASK-37, BR-TASK-08, BR-21
FR-TASK-47	The system shall stop deadline reminders for a task once it is Completed or Cancelled.	Must	–	REQ-TASK-38, BR-13

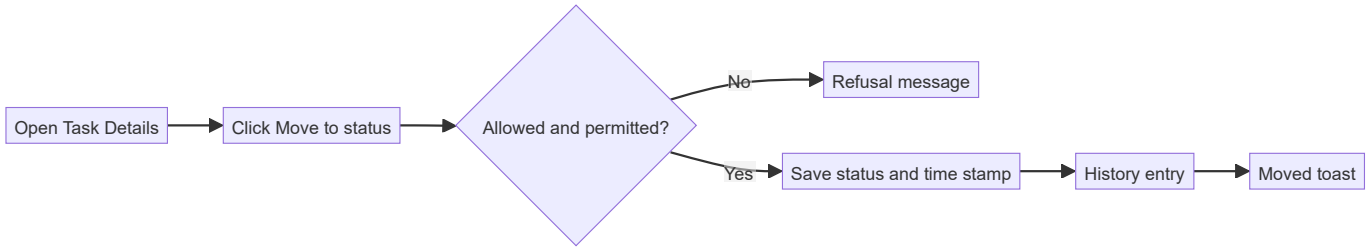
Use case: UC-TASK-05 – Edit a task

Item	Detail
Actor	Manager or Administrator
Preconditions	Holds `tasks.update`; the task is in the user's scope
Trigger	Clicks **Edit task**
Main flow	1. System shows the edit form with the current values. 2. User changes fields and clicks **Save changes**. 3. System validates and saves. 4. System records deadline, priority and worked-hours changes in the history. 5. If worked hours changed, the assignee and recipients are notified and mailed. 6. System shows "Task updated".
Alternate flows	A1: **Cancel** – nothing is saved.
Exceptions	E1: Invalid value → field message. E2: No permission → "Your role is not allowed to edit tasks." E3: Out of scope → "You do not have access to this task."
Postconditions	Task updated; changes recorded

Use case: UC-TASK-06 – Change a task's status

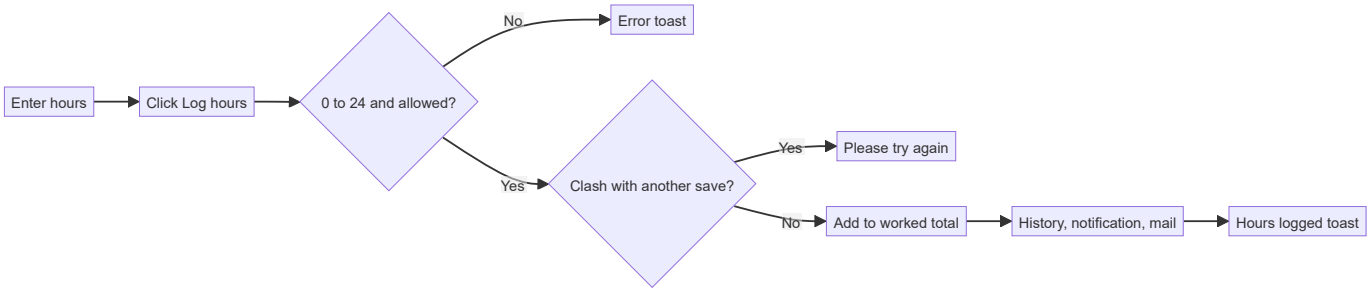
Item	Detail
Actor	Assignee, Manager or Administrator
Preconditions	Holds `tasks.status.change`; the task is in the user's scope
Trigger	Clicks a status under **Move to**
Main flow	1. System shows only the allowed next statuses. 2. User clicks one. 3. System checks the permission and the transition table. 4. System saves the status (and completion or cancel time), records the history entry and shows "{code} moved to {status}".

Item	Detail
Alternate flows	A1: Same as the Board drag (see `../02.3-Board/FRD-Board.md`).
Exceptions	E1: Disallowed move → "Cannot move a task from Completed to InProgress." (statuses vary). E2: No `tasks.status.change` → refused.
Postconditions	New status saved and recorded; reminders stop for Completed or Cancelled



Use case: UC-TASK-07 – Log worked hours

Item	Detail
Actor	Assignee (or a `tasks.update` holder)
Preconditions	Can access the task
Trigger	Enters hours in **Log hours** and confirms
Main flow	1. System checks 0 < hours ≤ 24 and that the user is the assignee or holds `tasks.update`. 2. System adds the hours to the worked total and records "Worked hours logged: {X}h (total)". 3. System notifies the assignee and recipients in-app and by mail ("… added 3h of worked time (was 2h) …"). 4. System shows "3h logged" and updates Worked and Remaining.
Alternate flows	–
Exceptions	E1: Hours empty or 0 → "Enter the hours worked (greater than 0)." E2: Outside 0–24 → "Hours must be greater than 0 and at most 24 per entry." E3: Not allowed → "Only the assignee can log hours on this task." E4: Clashing save → "The task's worked hours changed while you were saving. Please try again."
Postconditions	Worked total increased; history and notifications written



Use case: UC-TASK-08 – Delete a task

Item	Detail
Actor	Administrator
Preconditions	Holds `tasks.delete`
Trigger	Clicks **Delete** on a list row or on Task Details
Main flow	1. System asks "Delete task?" and explains the task will be removed for everyone while comments and history are kept. 2. User confirms. 3. System soft-deletes the task, records "Task deleted" and shows "{code} deleted".
Alternate flows	A1: User cancels the popup – nothing changes.
Exceptions	E1: No permission – the button is hidden and a direct request is refused.
Postconditions	Task hidden from every list, board, workload count and reminder; comments and history kept

Field validations

Field	Rule	Error message
Title (edit)	Required; at most 200 characters	"Title is required." · "Title must be 200 characters or fewer."
Deadline (edit)	Later than Start date	"Deadline must be after the start date."
Estimated hours (edit)	At least 0.5	"Enter at least 0.5 hours."
Log hours	Greater than 0, at most 24	"Enter the hours worked (greater than 0)." · "Hours must be greater than 0 and at most 24 per entry."

Business rules applied: BR-TASK-01, BR-TASK-03, BR-TASK-04, BR-TASK-07, BR-TASK-08, BR-TASK-12; global BR-10, BR-11, BR-12, BR-13, BR-17, BR-19, BR-21.

Notifications / side effects

Event	Who is notified	Channel
Worked hours changed (N-TASK-05, N-TASK-06)	Assignee and recipients	In-app, internal Mail
Status changed, task edited (other fields), task deleted	Nobody – recorded in the history only	–
Completed or Cancelled	Reminder engine stops further reminders	–

Messages

Situation	Message
Edit saved	"Task updated"
Status changed	"TSK-000123 moved to InProgress"
Hours logged	"3h logged"
Deleted	"TSK-000123 deleted"
Delete confirmation	"Delete task?" – "… will be removed for everyone. Its comments and history are kept."
Disallowed status move	"Cannot move a task from Completed to InProgress."
Not allowed to log hours	"Only the assignee can log hours on this task."

Situation	Message
Concurrent hours save	"The task's worked hours changed while you were saving. Please try again."
No access / permission	"You do not have access to this task." · "Your role is not allowed to edit tasks." · screen fallback "Could not load the task."

4.1.4 Screen: Reassign Task

Field	Value
Route / location	`/tasks/{id}/reassign`
Access	`tasks.reassign` (Administrator, Manager; Custom role as granted) and access to the task; the people list needs `employees.view`
Entry points	Task Details → **Reassign task** tab → **Choose new assignee**

Screen elements

Element	Type	Description	Mandatory	Visible to
"1 · Who takes over?"	Step	Current and New cards, "Search people…", list of active users except the current assignee	Yes	`tasks.reassign`
"2 · Why is it moving?" – Reason	Text area	Up to 500 characters; hint: "recorded in the activity history and shown to everyone notified"	Yes	`tasks.reassign`
"What happens next"	List	Everyone who will be told: new assignee, previous assignee, also notified	–	`tasks.reassign`
Cancel, Reassign	Buttons	Return to the task / start the reassignment	–	`tasks.reassign`
Confirmation popup	Dialog	"Reassign task?" – "… will move from A to B. The new assignee will be notified."	–	`tasks.reassign`

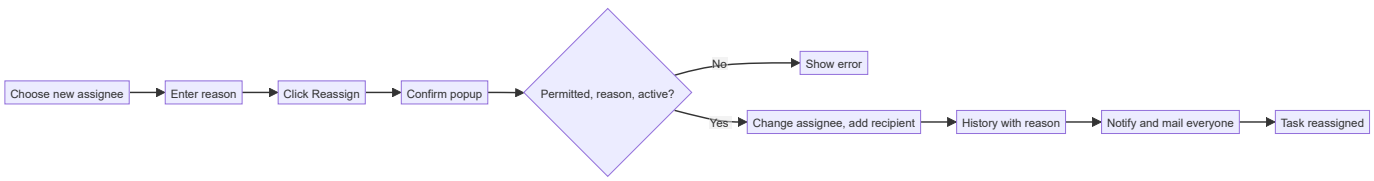
Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-48	The system shall let the user search the people list and pick any active user except the current assignee as the new assignee.	Must	`tasks.reassign`, `employees.view`	REQ-TASK-39
FR-TASK-49	The system shall refuse a reassignment by a user without `tasks.reassign` ("Your role is not allowed to reassign tasks.") or outside the task's scope ("You do not have access to this task.").	Must	`tasks.reassign`	REQ-TASK-39, BR-TASK-01, BR-10
FR-TASK-50	The system shall require a reason of at most 500 characters and refuse an empty one with "A reassignment reason is required.".	Must	`tasks.reassign`	REQ-TASK-40, BR-TASK-09, BR-18
FR-TASK-51	The system shall refuse a new assignee who does not exist or is not active with "The target employee does not exist or is not active.".	Must	`tasks.reassign`	REQ-TASK-39, BR-TASK-09

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-52	The system shall ask "Reassign task?" and save only after the user confirms.	Must	`tasks.reassign`	REQ-TASK-41
FR-TASK-53	The system shall record "Reassigned from {A} to {B}. Reason: {reason}" with the acting user and the time.	Must	`tasks.reassign`	REQ-TASK-42, BR-19
FR-TASK-54	The system shall add the new assignee as a recipient and keep all existing comments, recipients and history.	Must	`tasks.reassign`	REQ-TASK-43, BR-TASK-05, BR-18
FR-TASK-55	The system shall notify the new assignee, the previous assignee, the recipients and the acting user in-app and by internal mail, then show "Task reassigned" and return to the task.	Must	`tasks.reassign`	REQ-TASK-44
FR-TASK-56	The system shall list under "What happens next", before saving, everyone who will be told about the reassignment.	Could	`tasks.reassign`	REQ-TASK-45

Use case: UC-TASK-09 – Reassign a task

Item	Detail
Actor	Manager or Administrator
Preconditions	Holds `tasks.reassign`; the task is in the user's scope
Trigger	Task Details → **Reassign task** → **Choose new assignee**
Main flow	1. System shows the current assignee and the people list. 2. User searches and picks the new person. 3. User enters a reason; "What happens next" lists everyone who will be told. 4. User clicks **Reassign**; system asks "Reassign task?". 5. User confirms. 6. System checks permission, reason and the person's status, changes the assignee, adds them as a recipient and records the reason. 7. System notifies and mails everyone listed, shows "Task reassigned" and returns to Task Details.
Alternate flows	A1: **Cancel** or cancelling the popup – nothing changes.
Exceptions	E1: Reason empty → "A reassignment reason is required." E2: Target inactive → "The target employee does not exist or is not active." E3: No permission → "Your role is not allowed to reassign tasks."; without the key the route sends the user back to the dashboard.
Postconditions	New assignee set; history and comments kept; everyone informed



Field validations

Field	Rule	Error message
New assignee	Required; active user; not the current assignee	"The target employee does not exist or is not active."
Reason	Required; at most 500 characters	"A reassignment reason is required."

Business rules applied: BR-TASK-01, BR-TASK-05, BR-TASK-09, BR-TASK-12; global BR-10, BR-18, BR-19, BR-23.

Notifications / side effects

Event	Who is notified	Channel
Task reassigned (N-TASK-02, N-TASK-03, N-TASK-04)	New assignee, previous assignee, recipients, acting user	In-app, internal Mail

Messages

Situation	Message
Confirmation	"Reassign task?" – "… will move from A to B. The new assignee will be notified."
Success	"Task reassigned"
No permission	"Your role is not allowed to reassign tasks."

4.1.5 Screen: Task Discussion

Field	Value
Route / location	`/tasks/{id}/discussion` (summary in the Task Details **Discussion** tab)
Access	Users who can access the task; writing needs `tasks.comment.create`, `tasks.comment.edit`, `tasks.comment.delete`
Entry points	Task Details → **Discussion** → **Open full discussion** →

The full page (`TaskDiscussionComponent`, `discussion/pages/task-discussion.component.*`) has **Back to task #N**, chips with the task code, title and comment count, the thread, an **Add a comment** box and a side rail.

Screen elements

Element	Type	Description	Mandatory	Visible to
Header chips	Chips	Task code, title and "N comments"	–	Task access
Comment history	List	Grouped by day; author initials and name, "You" on the user's own comments, time, "(edited)" mark; oldest first. Empty state "No comments yet"	–	Task access
Comment box	Text area	**Add a comment**, at most 1000 characters on the page (the server accepts 4000); counter "N / 1000 · Everyone on the task is notified"; **Post comment**	Yes (to post)	`tasks.comment.create`
Edit / Delete	Actions	On the user's own comments: **Edit** opens the text in place with **Cancel** / **Save**; **Delete** asks "Delete comment?" first	–	author with `tasks.comment.edit` / author with `tasks.comment.delete`
Who is notified	Panel	The assignee and each recipient	–	Task access

Element	Type	Description	Mandatory	Visible to
Task	Panel	Status, priority and deadline	–	Task access

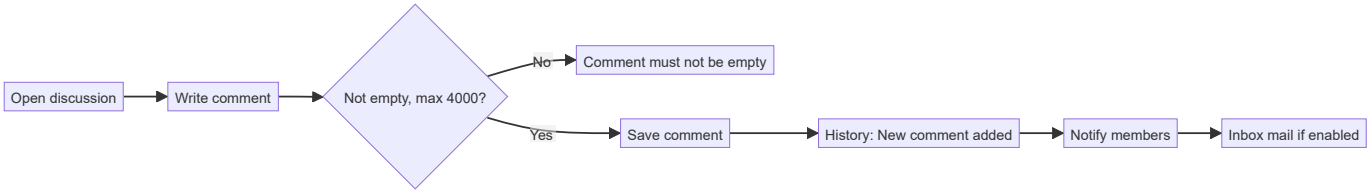
Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-57	The system shall list a task's comments oldest first, each with author, time and an "(edited)" mark when edited, and show "No comments yet" when there are none.	Must	task access	REQ-TASK-46
FR-TASK-58	The system shall let users with `tasks.comment.create` post a comment that is not empty and at most 4000 characters.	Must	`tasks.comment.create`	REQ-TASK-47
FR-TASK-59	The system shall let only the author edit (with `tasks.comment.edit`) or delete (with `tasks.comment.delete`) a comment, whatever other permissions the user holds.	Should	`tasks.comment.edit`, `tasks.comment.delete`	REQ-TASK-48, BR-TASK-10
FR-TASK-60	The system shall hide a deleted comment and keep an excerpt of up to 200 characters in the history.	Should	`tasks.comment.delete`	REQ-TASK-49, BR-21
FR-TASK-61	The system shall record "New comment added", "Comment edited" or "Comment deleted" in the task's history.	Must	–	REQ-TASK-50, BR-19
FR-TASK-62	The system shall notify the assignee, every recipient and the author in-app of each comment change, and place a system mail "Comment {added/edited/deleted}: {task}" in their Inbox when mail is enabled.	Should	–	REQ-TASK-51, BR-TASK-12

Use case: UC-TASK-10 – Post a comment

Item	Detail
Actor	Any user with task access and `tasks.comment.create`
Preconditions	Can access the task
Trigger	Writes in the comment box and posts
Main flow	1. System checks the comment is not empty and at most 4000 characters. 2. System saves it and records "New comment added". 3. System notifies the assignee, recipients and author ("{name} added a comment on "{task}": "{excerpt}"") and, when mail is enabled, places "Comment added: {task}" in their Inbox. 4. The comment appears at the end of the list.
Alternate flows	–
Exceptions	E1: Empty → "Comment must not be empty." E2: Task missing → "Task not found."

Item	Detail
Postconditions	Comment saved; history entry; members informed



Use case: UC-TASK-11 – Edit or delete an own comment

Item	Detail
Actor	Comment author
Preconditions	Holds `tasks.comment.edit` (edit) or `tasks.comment.delete` (delete)
Trigger	Chooses edit or delete on one of their comments
Main flow	1. System checks the user is the author. 2. Edit: saves the new text, marks it "(edited)", records "Comment edited". Delete: hides the comment, records "Comment deleted" with an excerpt. 3. System notifies members as in UC-TASK-10 ("edited" / "deleted").
Alternate flows	–
Exceptions	E1: Not the author → "Only the author can edit this comment." / "Only the author can delete this comment." E2: Empty edit → "Comment must not be empty."
Postconditions	Comment changed or hidden; history and notifications written

Field validations

Field	Rule	Error message
Comment	Required; at most 4000 characters	"Comment must not be empty."

Business rules applied: BR-TASK-10, BR-TASK-12; global BR-19, BR-21, BR-23.

Notifications / side effects

Event	Who is notified	Channel
Comment added, edited or deleted (N-TASK-07, N-TASK-08)	Assignee, recipients, author	In-app; system mail in the Inbox when mail is enabled

Messages

Situation	Message
Not the author	"Only the author can edit this comment." · "Only the author can delete this comment."
Task missing	"Task not found."
No comments	"No comments yet"

4.1.6 Screen: Activity History

Field	Value
Route / location	`/tasks/{id}/activity` (summary in the Task Details **Activity history** tab)
Access	`tasks.activity.view` and access to the task; worked-hours correction: Administrator role only
Entry points	Task Details → **Activity history** → **Open full history** →; Time tracking → "View hours in activity history"

The full page (`ActivityComponent` , `activity/pages/activity.component.*`) has **Back to task #N**, the title "Activity history", an event count, the filters and the entries, newest first and grouped by day.

Screen elements

Element	Type	Description	Mandatory	Visible to
Entries	List	Newest first, grouped by day; each entry has an event-type chip, time, message, old → new value and the acting user. Empty state "No activity found"	–	`tasks.activity.view`
Filters	Inputs	Event-type pills (**All**, Status changed, Comment added, Hours logged, Reassigned, Deadline changed) and an **Other events** select; **Search activity…**; **From** / **To** dates; **Clear** / **Apply**. The page has no person filter; the API accepts one (`actorUserId`)	No	`tasks.activity.view`
Edit hours / Delete	Actions	On "Worked hours logged" entries: **Edit hours** (min 0.25, step 0.25) or **Delete** (after confirmation). The page checks the role name Administrator, not a permission key	–	Administrator role

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-63	The system shall record history entries automatically for task created, assigned, reassigned, status changed, deadline changed, priority changed, completed, comment added, edited and deleted, worked hours logged, recipients updated ("Task recipients updated"), task deleted and administrator corrections.	Must	–	REQ-TASK-52, BR-19
FR-TASK-64	The system shall show the entries in chronological order with the acting user, old value, new value and time, and show "No activity yet" when there are none.	Must	`tasks.activity.view`	REQ-TASK-53
FR-TASK-65	The system shall filter entries by event type, person, from/to date (a date without time covers the whole day) and message text.	Should	`tasks.activity.view`	REQ-TASK-54
FR-TASK-66	The system shall show the history (tab and page) only to users with `tasks.activity.view` who can access the task, hide the tab otherwise and refuse a direct request.	Must	`tasks.activity.view`	REQ-TASK-55
FR-TASK-67	The system shall offer no way to change or delete history entries, except the Administrator correction in FR-TASK-68.	Must	–	REQ-TASK-56, BR-TASK-11, BR-19

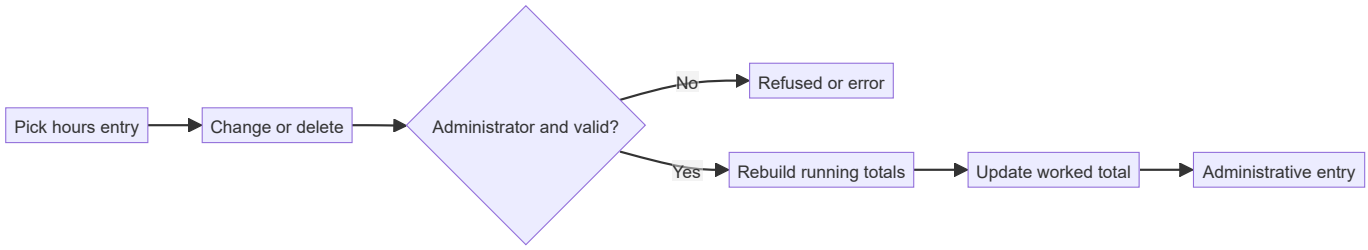
ID	Requirement	Priority	Permission	BRD trace
FR-TASK-68	The system shall let the Administrator role change the hours of a "Worked hours logged" entry (more than 0, at most 24) or delete it, recalculate every later entry and the task's worked total, and record an "Administrative" entry such as "Administrator changed a worked-hours entry from 3h to 2h; worked total is now 5h".	Could	Administrator role	REQ-TASK-57, BR-TASK-11
FR-TASK-69	The system shall open the history filtered to worked-hours entries from the Time tracking link "View hours in activity history".	Could	`tasks.activity.view`	REQ-TASK-58

Use case: UC-TASK-12 – Review and filter the history

Item	Detail
Actor	Any user with `tasks.activity.view`
Preconditions	Can access the task
Trigger	Opens **Activity history** or **Open full history** →
Main flow	1. System lists the entries in time order. 2. User filters by event type, person, date range or text. 3. System shows the matching entries with old and new values.
Alternate flows	A1: Opened from "View hours in activity history" – pre-filtered to worked-hours entries.
Exceptions	E1: No `tasks.activity.view` → tab hidden, direct request refused.
Postconditions	Nothing changed

Use case: UC-TASK-13 – Correct a worked-hours entry

Item	Detail
Actor	Administrator
Preconditions	Administrator role; the task has a "Worked hours logged" entry
Trigger	Chooses to change or remove that entry
Main flow	1. Administrator enters the corrected hours or chooses delete. 2. System checks the role and the hours. 3. System recalculates the running totals and the task's worked total. 4. System records the "Administrative" entry.
Alternate flows	A1: Delete instead of change – the entry's hours are removed from the total.
Exceptions	E1: Hours outside range → "Hours must be greater than 0 and at most 24 per entry." E2: Entry added no hours → "This entry did not add hours, so it can only be deleted." E3: Not Administrator → refused.
Postconditions	Worked total corrected; the correction itself is on record



Field validations

Field	Rule	Error message
Corrected hours	Greater than 0, at most 24	"Hours must be greater than 0 and at most 24 per entry."
Entry type	Only an entry that added hours can be changed	"This entry did not add hours, so it can only be deleted."
From / to date (filter)	A date without time covers the whole day	–

Business rules applied: BR-TASK-11; global BR-19.

Notifications / side effects

Event	Who is notified	Channel
Administrator correction	Nobody – recorded as an "Administrative" entry	–

Messages

Situation	Message
No entries	"No activity yet"
Bad correction	"Hours must be greater than 0 and at most 24 per entry." · "This entry did not add hours, so it can only be deleted."
No permission	Tab hidden; a direct request is refused

4.1.7 Common behaviour (all Task Management screens)

Field	Value
Route / location	All routes in 4.1.1–4.1.6
Access	As per screen
Entry points	–

Functional requirements

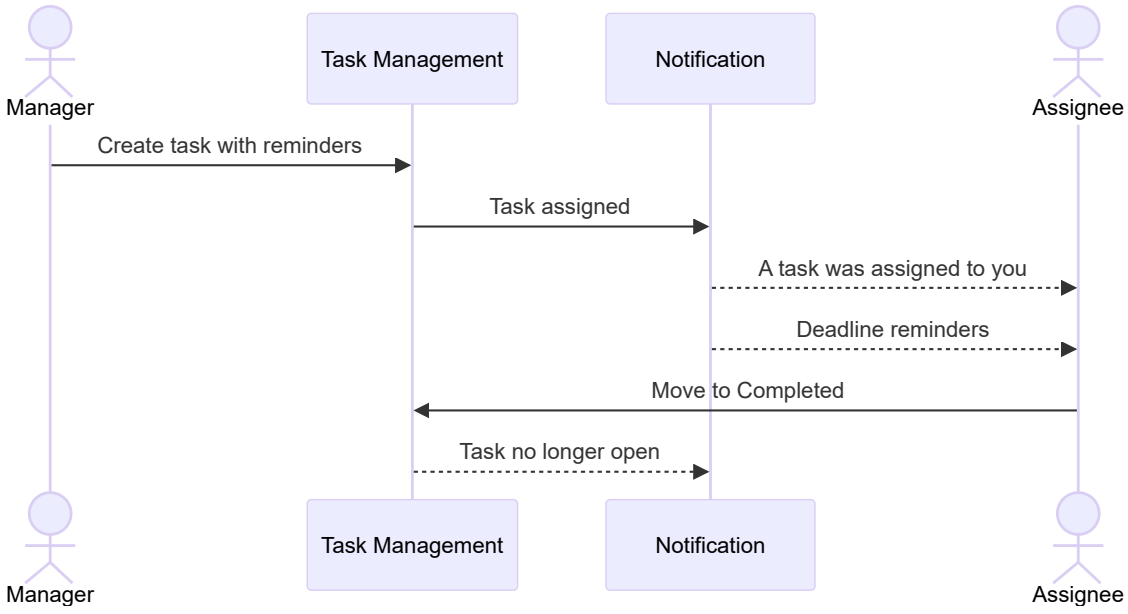
ID	Requirement	Priority	Permission	BRD trace
FR-TASK-70	The system shall never notify deleted users, skip in-app notifications when they are switched off in Application Settings, and skip comment mail when mail is switched off.	Must	–	BR-TASK-12
FR-TASK-71	The system shall keep a task change when a notification or mail for it fails, and log the failure.	Must	–	REQ-TASK-34, REQ-TASK-44, REQ-TASK-51, NFR-TASK-04
FR-TASK-72	The system shall check the task scope and the action permission on the server for every action, so hiding a button is never the only protection.	Must	all task keys	REQ-TASK-09, NFR-TASK-02, BR-04

ID	Requirement	Priority	Permission	BRD trace
FR-TASK-73	The system shall send a user without the route's permission (Create Task, Reassign Task, Activity History) back to their dashboard.	Must	`tasks.create`, `tasks.reassign`, `tasks.activity.view`	REQ-TASK-09, REQ-TASK-55

5. Cross-Module Workflows

5.1 Create a task → notify → remind

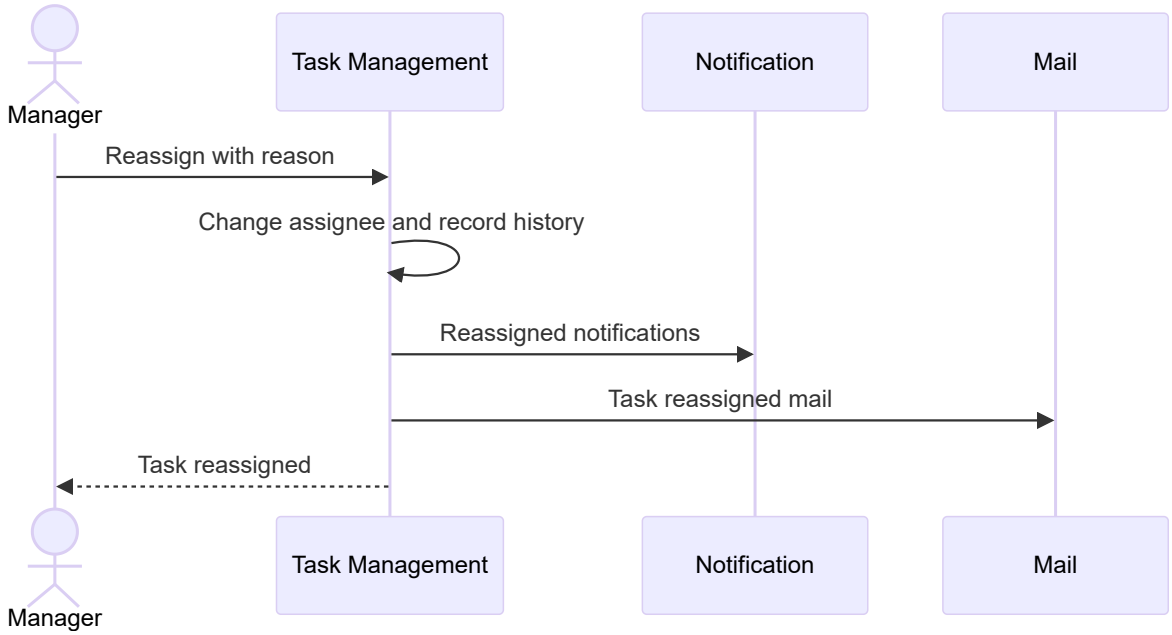
Step	Module	Actor / System	Action	FR reference
1	TASK	Manager	Creates and assigns a task with reminder intervals	FR-TASK-14…FR-TASK-26
2	NOTIF	System	Assignee receives "A task was assigned to you: …"	FR-TASK-27
3	NOTIF	Reminder engine	Sends first, second, due-today and overdue reminders from the task's intervals	Notification FRD
4	TASK	Assignee	Moves the task to Completed	FR-TASK-37…FR-TASK-40
5	NOTIF	Reminder engine	Sends no further reminders	FR-TASK-47



5.2 Reassign a task → notifications and internal mail

Step	Module	Actor / System	Action	FR reference
1	WORK	Manager	Spots a busy employee and opens their tasks (**Tasks** →)	FR-WORK-15 (Workload FRD)
2	TASK	Manager	Reassigns a task with a reason	FR-TASK-48…FR-TASK-54

Step	Module	Actor / System	Action	FR reference
3	NOTIF	System	In-app notifications to new and previous assignee, recipients and actor	FR-TASK-55
4	MAIL	System	Internal mail "Task reassigned: {title}" to the same people	FR-TASK-55



5.3 Comment → notification and Inbox mail

Step	Module	Actor / System	Action	FR reference
1	TASK	User	Posts, edits or deletes a comment	FR-TASK-57…FR-TASK-61
2	NOTIF	System	In-app notification to assignee, recipients and author	FR-TASK-62
3	MAIL	System	System mail "Comment added: {task}" in each Inbox when mail is enabled	FR-TASK-62, FR-TASK-70

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6. Reports & Exports

Not applicable for this sub-module. Task reports and CSV exports are specified in `../../05-Reports/05.1-Reports/FRD-Reports.md`.

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7. Notifications & Messages Catalogue

Code	Trigger	Recipients	Channel	Text
N-TASK-01	Task created	Assignee	In-app (Task assigned)	"A task was assigned to you: {title}"

Code	Trigger	Recipients	Channel	Text
N-TASK-02	Task reassigned	New assignee	In-app (Reassigned)	"A task was reassigned to you: {title}. Reason: {reason}"
N-TASK-03	Task reassigned	Previous assignee, recipients, acting user	In-app (Reassigned)	"{actor} reassigned task "{title}" ({code}) from {A} to {B}. Reason: {reason}"
N-TASK-04	Task reassigned	New and previous assignee, recipients, acting user	Internal Mail	Subject "Task reassigned: {title}"
N-TASK-05	Worked hours changed (Log hours or edit)	Assignee, recipients	In-app (System)	"{actor} added {X}h of worked time (was {Y}h) on task {title}. Total worked now: …h of …h estimated (…h remaining | over by …h)." – wording becomes "cleared the worked hours" or "updated the worked hours" for those cases
N-TASK-06	Worked hours changed	Assignee, recipients	Internal Mail	Subject "Worked hours changed: {title}"
N-TASK-07	Comment added, edited or deleted	Assignee, recipients, author	In-app (Task comment)	"{name} added a comment on "{task}": "{excerpt}"" (also "edited", "deleted")
N-TASK-08	Comment added, edited or deleted	Assignee, recipients, author	System mail in the Inbox (when mail is enabled)	Subject "Comment added: {task}" / "Comment edited: {task}" / "Comment deleted: {task}"

Status changes, edits of other fields and deletes send no notification. Deadline reminders are catalogued in the Notification FRD.

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8. Data Requirements (functional view)

Data item	Description	Created by	Rules
Task code	Readable unique identifier	System	`TSK-` + 6-digit number; never reused
Task	Project, title, description, assignee, creator, priority, start date, deadline, estimated hours, worked hours, status, completed / cancelled time, reminder intervals, created / updated time	Manager / Administrator	Starts as To Do; soft delete only; reminder intervals fixed after creation
Task recipient	A user notified about a task	System (assignee) / creator (additional)	Assignee always included; at most 50 additional; must exist

Data item	Description	Created by	Rules
Task comment	Discussion text with author, time and edited mark	Any user with `tasks.comment.create`	Not empty; at most 4000 characters; soft delete
Activity history entry	Event type, message, old value, new value, acting user, time	System	Never changed by users except Administrator worked-hours corrections, which are themselves recorded
Stage	Upcoming, Due today, Overdue, Completed	System (on display)	Worked out in the application time zone; never stored
Remaining hours	Estimated minus worked	System (on display)	Shows "over by" when worked exceeds estimated

9. Non-Functional Requirements (functional impact)

ID	Requirement	Affects
NFR-TASK-01	20 tasks per page by default, at most 100; search refreshes 300 ms after typing stops	Task List
NFR-TASK-02	Task scope and every action permission checked on the server	All screens (FR-TASK-72)
NFR-TASK-03	Delete and Reassign always ask for confirmation; validation messages appear next to the field	Task Details, Reassign Task, Create Task
NFR-TASK-04	A failed notification or mail never undoes the task change	Create, Reassign, Log hours, Discussion (FR-TASK-71)
NFR-TASK-05	Simultaneous worked-hours saves never lose hours	Task Details (FR-TASK-44)
NFR-TASK-06	One status transition table shared by Task Details and the Board	Task Details, Board
NFR-TASK-07 / NFR-TASK-08	Compatibility and accessibility as the global NFRs (NFR-16, NFR-18)	All screens

10. Assumptions & Open Issues

Type	Description	Owner
Assumption	Managers' teams are maintained in Management (department managers, project-manager flags)	Jojo George
Assumption	Creators of tasks also hold `employees.view`, `projects.view` and `settings.reminders.manage` (true for Administrator and Manager by default)	Jojo George
Open issue	Delete is not checked against task scope: any `tasks.delete` holder can delete any task (R-TASK-01)	Jojo George
Open issue	A Manager can read and write comments and history of another team's task by direct request; the team limit applies on screen only (R-TASK-02)	Jojo George
Open issue	The **Move to** buttons are shown without `tasks.status.change`; the server then refuses (R-TASK-03)	Jojo George

Type	Description	Owner
Open issue	Only Employee-role users can be chosen as assignee on Create; custom-role users and Managers only through Reassign (R-TASK-04)	Jojo George
Open issue	Reminder intervals and additional recipients cannot be changed after creation; `tasks.recipients.manage` has no screen (R-TASK-05)	Jojo George
Open issue	Clearing Worked hours in the edit form records a change to 0 (R-TASK-06)	Jojo George
Open issue	For a Manager, picking a Project in the normal list filter also drops the Manager's own non-team tasks, because the project filter is handled like a drill-down	Jojo George
Open issue	The Task List **Board view** button is shown without `board.view`; the Board then sends the user back	Jojo George

11. Requirements Traceability Matrix

Test cases name the automated test files (`TaskFlow.UnitTests` / `TaskFlow.IntegrationTests` and the frontend `*.spec.ts` files under `frontend/src/app/`) that test the behaviour; the mapping was checked against the test sources.

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-TASK-01	FR-TASK-01	Task List	UC-TASK-01	`TasksEndpointsTests.cs`, `TaskServiceTests.cs`, `TaskManagementServiceTests.cs`
REQ-TASK-02	FR-TASK-03, FR-TASK-13	Task List	UC-TASK-01	`TasksEndpointsTests.cs`, `task-list.component.spec.ts`
REQ-TASK-03	FR-TASK-04	Task List	UC-TASK-01	`TaskManagementServiceTests.cs`, `TasksEndpointsTests.cs`, `task-list.component.spec.ts`
REQ-TASK-04	FR-TASK-05, FR-TASK-13	Task List	UC-TASK-01	`TaskManagementServiceTests.cs`, `TasksEndpointsTests.cs`, `ProjectTaskTests.cs`, `task-list.component.spec.ts`
REQ-TASK-05	FR-TASK-06	Task List	UC-TASK-01	`task-list.component.spec.ts`
REQ-TASK-06	FR-TASK-07	Task List	UC-TASK-01	`TaskTrackingDomainTests.cs`, `TaskManagementServiceTests.cs`, `task-list.component.spec.ts`
REQ-TASK-07	FR-TASK-08	Task List	UC-TASK-02	`AnalyticsDrillDownTests.cs`, `TaskBucketRulesTests.cs`, `task-list.component.spec.ts`
REQ-TASK-08	FR-TASK-10	Task List	UC-TASK-01	`task-list.component.spec.ts`
REQ-TASK-09	FR-TASK-11, FR-TASK-72, FR-TASK-73	Task List	UC-TASK-03, UC-TASK-08	`CustomRolePermissionTests.cs`, `task-list.component.spec.ts`, `app.routes.spec.ts`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-TASK-10	FR-TASK-12	Task List	UC-TASK-01	`task-list.component.spec.ts`
REQ-TASK-11	FR-TASK-14, FR-TASK-28	Create Task	UC-TASK-03	`TaskManagementServiceTests.cs`, `TasksEndpointsTests.cs`, `task-create.component.spec.ts`
REQ-TASK-12	FR-TASK-15	Create Task	UC-TASK-03	`CreateTaskRequestValidatorTests.cs`, `ProjectTaskTests.cs`, `task-create.component.spec.ts`
REQ-TASK-13	FR-TASK-16	Create Task	UC-TASK-03	`task-create.component.spec.ts` (title length only; no test for the 4000-character description)
REQ-TASK-14	FR-TASK-17	Create Task	UC-TASK-03	`CreateTaskRequestValidatorTests.cs`, `task-create.component.spec.ts`
REQ-TASK-15	FR-TASK-18	Create Task	UC-TASK-03	`task-create.component.spec.ts`
REQ-TASK-16	FR-TASK-19, FR-TASK-35	Create Task, Task Details	UC-TASK-03, UC-TASK-05	`CreateTaskRequestValidatorTests.cs`, `TaskValidatorsTests.cs`, `task-create.component.spec.ts`, `task-details.component.spec.ts`
REQ-TASK-17	FR-TASK-20	Create Task	UC-TASK-03	`CreateTaskRequestValidatorTests.cs`, `task-create.component.spec.ts`
REQ-TASK-18	FR-TASK-21	Create Task	UC-TASK-03	`TaskManagementServiceTests.cs`
REQ-TASK-19	FR-TASK-22	Create Task	UC-TASK-03	`CreateTaskRequestValidatorTests.cs`, `TaskManagementServiceTests.cs`, `task-create.component.spec.ts`
REQ-TASK-20	FR-TASK-23	Create Task	UC-TASK-03	`CreateTaskRequestValidatorTests.cs`, `TaskManagementServiceTests.cs`, `TasksEndpointsTests.cs`, `task-create.component.spec.ts`
REQ-TASK-21	FR-TASK-24	Create Task	UC-TASK-03	`ProjectTaskTests.cs`, `TaskManagementServiceTests.cs`
REQ-TASK-22	FR-TASK-25	Create Task	UC-TASK-03	`task-create.component.spec.ts`
REQ-TASK-23	FR-TASK-26	Create Task	UC-TASK-03	`TaskManagementServiceTests.cs`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-TASK-24	FR-TASK-27, FR-TASK-28	Create Task	UC-TASK-03	`TaskManagementServiceTests.cs`, `task-create.component.spec.ts`
REQ-TASK-25	FR-TASK-29, FR-TASK-30	Create Task	UC-TASK-04	`task-create.component.spec.ts`
REQ-TASK-26	FR-TASK-31, FR-TASK-32	Task Details	UC-TASK-05	`TasksEndpointsTests.cs`, `TaskServiceTests.cs`, `task-details.component.spec.ts`
REQ-TASK-27	FR-TASK-33	Task Details	UC-TASK-06	`TaskTrackingDomainTests.cs`
REQ-TASK-28	FR-TASK-34, FR-TASK-35	Task Details	UC-TASK-05	`TaskValidatorsTests.cs`, `TaskServiceTests.cs`, `TasksEndpointsTests.cs`, `task-details.component.spec.ts`
REQ-TASK-29	FR-TASK-36	Task Details	UC-TASK-05	`TaskServiceTests.cs`, `TasksEndpointsTests.cs`
REQ-TASK-30	FR-TASK-37, FR-TASK-38, FR-TASK-39	Task Details	UC-TASK-06	`TaskTrackingDomainTests.cs`, `TaskServiceTests.cs`, `TasksEndpointsTests.cs`, `task-details.component.spec.ts`
REQ-TASK-31	FR-TASK-40	Task Details	UC-TASK-06	`TaskServiceTests.cs` (completion time only; no test for the cancel time)
REQ-TASK-32	FR-TASK-41	Task Details	UC-TASK-07	`TaskLogHoursTests.cs`, `task-details.component.spec.ts`
REQ-TASK-33	FR-TASK-42	Task Details	UC-TASK-07	`task-details.component.spec.ts`
REQ-TASK-34	FR-TASK-43, FR-TASK-71	Task Details	UC-TASK-07	`TaskWorkedHoursNotificationTests.cs`, `TaskLogHoursTests.cs`
REQ-TASK-35	FR-TASK-44	Task Details	UC-TASK-07	`TaskLogHoursTests.cs`
REQ-TASK-36	FR-TASK-45	Task Details	UC-TASK-12	`task-details.component.spec.ts`
REQ-TASK-37	FR-TASK-46	Task Details, Task List	UC-TASK-08	`TaskDeletionTests.cs`, `task-details.component.spec.ts`, `task-list.component.spec.ts`
REQ-TASK-38	FR-TASK-47	Task Details	UC-TASK-06	`ReminderServiceTests.cs`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-TASK-39	FR-TASK-48, FR-TASK-49, FR-TASK-51	Reassign Task	UC-TASK-09	`TaskServiceTests.cs`, `TasksEndpointsTests.cs`, `reassign.component.spec.ts`
REQ-TASK-40	FR-TASK-50	Reassign Task	UC-TASK-09	`TaskValidatorsTests.cs`, `reassign.component.spec.ts`
REQ-TASK-41	FR-TASK-52	Reassign Task	UC-TASK-09	`reassign.component.spec.ts`
REQ-TASK-42	FR-TASK-53	Reassign Task	UC-TASK-09	`TaskServiceTests.cs`
REQ-TASK-43	FR-TASK-54	Reassign Task	UC-TASK-09	`TaskServiceTests.cs`
REQ-TASK-44	FR-TASK-55, FR-TASK-71	Reassign Task	UC-TASK-09	`TaskServiceTests.cs`, `TasksEndpointsTests.cs`, `reassign.component.spec.ts`
REQ-TASK-45	FR-TASK-56	Reassign Task	UC-TASK-09	`reassign.component.spec.ts`
REQ-TASK-46	FR-TASK-57	Task Discussion	UC-TASK-10	`TaskCommentActivityNotificationTests.cs`, `TasksEndpointsTests.cs`, `task-discussion.component.spec.ts`
REQ-TASK-47	FR-TASK-58	Task Discussion	UC-TASK-10	`TaskCommentActivityNotificationTests.cs`, `TasksEndpointsTests.cs`, `task-discussion.component.spec.ts`
REQ-TASK-48	FR-TASK-59	Task Discussion	UC-TASK-11	`TaskCommentActivityNotificationTests.cs`, `task-discussion.component.spec.ts`
REQ-TASK-49	FR-TASK-60	Task Discussion	UC-TASK-11	`TaskCommentActivityNotificationTests.cs`, `task-discussion.component.spec.ts`
REQ-TASK-50	FR-TASK-61	Task Discussion	UC-TASK-10, UC-TASK-11	`TaskCommentActivityNotificationTests.cs`, `TasksEndpointsTests.cs`
REQ-TASK-51	FR-TASK-62, FR-TASK-71	Task Discussion	UC-TASK-10	`TaskCommentActivityNotificationTests.cs`
REQ-TASK-52	FR-TASK-39, FR-TASK-63	Activity History	UC-TASK-12	`TaskManagementServiceTests.cs`, `TaskServiceTests.cs`, `TaskCommentActivityNotificationTests.cs`
REQ-TASK-53	FR-TASK-64	Activity History	UC-TASK-12	`TasksEndpointsTests.cs`, `TaskCommentActivityNotificationTests.cs`, `activity.component.spec.ts`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-TASK-54	FR-TASK-65	Activity History	UC-TASK-12	`TasksEndpointsTests.cs`, `TaskCommentActivityNotificationTests.cs`, `activity.component.spec.ts`
REQ-TASK-55	FR-TASK-66, FR-TASK-73	Activity History	UC-TASK-12	`app.routes.spec.ts`
REQ-TASK-56	FR-TASK-67	Activity History	UC-TASK-12	No automated test (no endpoint changes history apart from FR-TASK-68)
REQ-TASK-57	FR-TASK-68	Activity History	UC-TASK-13	`TaskCommentActivityNotificationTests.cs`, `activity.component.spec.ts`
REQ-TASK-58	FR-TASK-69	Activity History	UC-TASK-12	`activity.component.spec.ts`
BR-TASK-01	FR-TASK-01, FR-TASK-02, FR-TASK-32, FR-TASK-49	Task List, Task Details, Reassign Task	UC-TASK-01, UC-TASK-09	`TaskServiceTests.cs`, `TasksEndpointsTests.cs`
BR-TASK-02	FR-TASK-09	Task List	UC-TASK-02	`ManagerDrillDownScopeTests.cs`
BR-TASK-03	FR-TASK-37, FR-TASK-38	Task Details	UC-TASK-06	`TaskTrackingDomainTests.cs`
BR-TASK-04	FR-TASK-33	Task List, Task Details	UC-TASK-01	`TaskTrackingDomainTests.cs`
BR-TASK-05	FR-TASK-22, FR-TASK-54	Create Task, Reassign Task	UC-TASK-03, UC-TASK-09	`CreateTaskRequestValidatorTests.cs`, `TaskServiceTests.cs`
BR-TASK-06	FR-TASK-23	Create Task	UC-TASK-03	`CreateTaskRequestValidatorTests.cs`
BR-TASK-07	FR-TASK-41	Task Details	UC-TASK-07	`TaskLogHoursTests.cs`
BR-TASK-08	FR-TASK-46	Task List, Task Details	UC-TASK-08	`TaskDeletionTests.cs`
BR-TASK-09	FR-TASK-50, FR-TASK-51	Reassign Task	UC-TASK-09	`TaskValidatorsTests.cs`, `TaskServiceTests.cs`
BR-TASK-10	FR-TASK-59	Task Discussion	UC-TASK-11	`TaskCommentActivityNotificationTests.cs`, `task-discussion.component.spec.ts`

BRD item	FRD requirement(s)	Screen	Use case	Test case
BR-TASK-11	FR-TASK-67, FR-TASK-68	Activity History	UC-TASK-13	`TaskCommentActivityNotificationTests.cs`, `activity.component.spec.ts`
BR-TASK-12	FR-TASK-27, FR-TASK-62, FR-TASK-70	Create Task, Task Discussion	UC-TASK-03, UC-TASK-10	`TaskManagementServiceTests.cs`, `TaskCommentActivityNotificationTests.cs`

12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	Jojo George	Pending	–

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